

Γ 1-form.

Γ Def. A map $Q: \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3 \rightarrow \mathbb{R}$ is called a 1-form if: for fixed $p \in \mathbb{R}^3$,

$$Q(a \cdot V_p + W_p) = a \cdot Q(V_p) + Q(W_p) \text{ for all } a \in \mathbb{R} \text{ and } V_p, W_p \in T_p \mathbb{R}^3.$$

That is, the restriction $Q_p: T_p \mathbb{R}^3 \rightarrow \mathbb{R}: V_p \mapsto Q(V_p)$ is a linear functional.

Γ Def. Let Q_1, Q_2 be 1-forms, and $f, g: \mathbb{R}^3 \rightarrow \mathbb{R}$ be functions, and $V, W: \mathbb{R}^3 \rightarrow \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3$ be vector fields. Define the operations:

$$(Q_1 + Q_2)(V_p) \stackrel{\text{def}}{=} Q_1(V_p) + Q_2(V_p) \text{ and } (fQ)(V_p) \stackrel{\text{def}}{=} f(p) \cdot Q(V_p)$$

$$Q(fV + gW) \stackrel{\text{def}}{=} fQ(V) + gQ(W) \text{ and } (fQ_1 + gQ_2)(V) = fQ_1(V) + gQ_2(V)$$

Γ Def. Given a differentiable function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, the 1-form

$$df: \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3 \rightarrow \mathbb{R}: V_p \mapsto V_p[f]$$

is called the differential df .

Further, for the projections $\chi_i: \mathbb{R}^3 \rightarrow \mathbb{R}: (x_1, x_2, x_3) \mapsto x_i$ ($i=1, 2, 3$).

the set $\{(dx_1)_p, (dx_2)_p, (dx_3)_p\}$ is a dual basis for $T_p \mathbb{R}^3$ corresponding to $\{(e_1)_p, (e_2)_p, (e_3)_p\}$ for each $p \in \mathbb{R}^3$.

Γ Lemma. Given 1-form Q on $\mathbb{R}^3$,

$$Q = \sum_{i=1}^3 Q \circ e_i \cdot dx_i.$$

In particular, $df = \sum_{i=1}^3 df \circ e_i \cdot dx_i = \sum_{i=1}^3 \frac{\partial f}{\partial x_i} \cdot dx_i$

Γ Example.

Let $f(x, y, z) = (x^2 - 1)y + (y^2 + 2)z$. Then,

$$\begin{aligned} df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz \\ &= 2xy dx + (x^2 + 2yz - 1) dy + (y^2 + 2) dz. \end{aligned}$$

So, $V_p[f] = df(V_p) = 2p_1 p_2 V_1 + (p_1^2 + 2p_2 p_3 - 1) V_2 + (p_2^2 + 2) V_3$.

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